



Thermodynamics Systems

A thermodynamic system refers to a matter enclosed within a defined volume. Everything under study is part of the system. Everything external to the system is defined as the **surroundings** or **environment**. The interface between the system and the environment is called the **boundary**. This is where most interaction of energy happens.

ADIABATIC SYSTEM

It is a system that has no exchange of energy or matter with surrounding environment.

OPEN SYSTEM

If mass flows through the system across system boundary. In an open system, heat, work and matter may cross the boundary. This is often called "**control volume system**". Examples: Jet engine, pumps

STEADY FLOW SYSTEM

It is a type of open system wherein matter enters and leaves at the same rate. Examples: Boilers, turbines *Excel Review Center*

CLOSED SYSTEM

If no mass flows through the system boundaries, i.e. matter does not cross boundary but energy can cross the boundary. This is often called "**control mass system**".

ISOLATED SYSTEM

It is a type of closed system wherein neither mass nor energy crosses the boundaries.

DIATHERMIC SYSTEM

If energy crosses the system boundaries.

Characteristics of the different thermodynamic systems: *Excel Review Center*

System	Characteristics
Adiabatic	Heat must not cross the boundary
Open	Heat, work and matter may cross the boundary
Closed	Matter may not cross the boundary
Isolated	Matter and energy may not cross the boundary
Diathermic	Heat may cross the boundary

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Thermodynamics Processes

A thermodynamic process is defined as the energetic evolution a thermodynamic system proceeding from an initial state to a final state.
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ADIABATIC PROCESS

A process where no heat or other processes crosses the system boundary. The term "**adiabatic**" comes from Greek word "adiabatos" which means "not to be passed".

ISOBARIC PROCESS

A process in which the pressure remains constant.

ISOTHERMAL PROCESS

A process in which the temperature remains constant.

ISOVOLUMETRIC PROCESS

A process in which the volume remains constant. Also known as isochoric or isometric process *Excel Review Center*

ISENTROPIC PROCESS

An adiabatic process in which there is no change in system entropy

THROTTLING PROCESS

An adiabatic process in which there is no change in system enthalpy but with significant drop in pressure. The ratio of the change in temperature to the change in pressure when a real gas is throttled is called **Joule-Thomson coefficient**. This coefficient is also known as the **Joule-Kelvin coefficient**.

QUASISTATIC PROCESS

A process which can be divided infinitesimally from equilibrium

POLYTROPIC PROCESS

A process that follows the polytropic equation of state

REVERSIBLE PROCESS

A process in which its conclusion, both the system and the local surroundings can be restored to their initial state

IRREVERSIBLE PROCESS

A process in which its conclusion, both the system and the local surroundings cannot be restored to their initial state. All actual processes are irreversible processes.

Thermodynamics Processes Formulas

ADIABATIC PROCESS

$$Q = 0$$

$$\Delta U = -W$$

ISOBARIC PROCESS

$$\Delta P = 0$$

$$Q = \Delta H$$

ISOTHERMAL PROCESS

$$\Delta T = 0$$

$$Q = W$$

ISOVOLUMETRIC PROCESS

$$\Delta V = 0$$

$$Q = \Delta U$$

$$W = 0$$

ISENTROPIC PROCESS

$$\Delta S = 0$$

$$Q = 0$$

THROTTLING PROCESS

$$\Delta H = 0$$

$$P_2 < P_1$$

POLYTROPIC PROCESS

$$P_1 V_1^n = P_2 V_2^n$$

where:
 n = polytropic exponent

$n = 0$ for constant pressure process

$n = 1$ for constant temperature process

$n = k$ for isentropic process

$n = \infty$ for constant volume process

$$Q = mc_n \Delta T$$

$$c_n = \frac{n - k}{n - 1} c_v$$

where:

c_n = polytropic specific heat

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